

Highlights

Transport-Inspired Rare-Event Sampling in Spin Systems: Learned Importance Coordinates Improve Access to Deep Targets

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- Adjoint importance and the rare-event committor share the same backward-conditioning structure for first-passage problems, and coincide for an absorbing target.
- Exact tests validate both the zero-variance importance limit and unbiased WE population control.
- At matched update cost, WE reaches a tail bin approximately 24 times below the naive sampling floor near 5.9×10^{-6} , where direct sampling returns no estimate at all.
- A learned importance coordinate significantly improves deep-event discovery in the Edwards–Anderson spin glass, pooled across three independent disorder realizations (70.0% versus 51.1%, $p = 0.014$) and, separately, across three trained networks on one realization (66.7% versus 48.9%, $p = 0.023$).
- The improvement is in reliability of access, not efficiency: directly measured inference cost leaves the learned coordinate at roughly one-third of parity with the energy coordinate in wall-clock figure of merit (0.33, 95% CI [0.19, 0.58]).

Transport-Inspired Rare-Event Sampling in Spin Systems: Learned Importance Coordinates Improve Access to Deep Targets

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ARTICLE INFO

Keywords:

Rare-event sampling
committor
adjoint importance
Weighted Ensemble
variance reduction
Ising model
spin glass
neural importance

ABSTRACT

This work transfers the variance-reduction toolkit of radiation transport, adjoint importance functions and weight-window population control, to rare-event sampling in lattice spin systems, using the shared backward-conditioning structure of the transport adjoint and the transition-path committor. Exact references validate the implementation: a birth–death process reproduces the zero-variance limit of exact importance, and enumeration of a 4×4 lattice confirms that weight-conserving Weighted Ensemble (WE) resampling is unbiased for any binning coordinate. At matched spin-update cost, WE reaches an $L = 20$ Ising magnetization-tail bin about 24 times below the direct-sampling floor of 5.9×10^{-6} . For the frustrated Edwards–Anderson spin glass, the weighted population is binned by a self-consistency-trained neural coordinate acting on the full spin configuration. At the deepest calibrated target at $L = 16$, pooled across three independent disorder realizations with one trained network each, the event is observed in 63/90 learned-coordinate replicas against 46/90 for the physical energy coordinate (Fisher exact $p = 0.014$); a complementary check pooling three independently trained networks on a single realization against a matched-size baseline gives 60/90 against 44/90 ($p = 0.023$). The gain is in reliability of access, not efficiency: repeated network inference costs four to five times the wall-clock time of the energy coordinate, leaving a directly measured, two-level-bootstrapped figure of merit at roughly one-third of parity (0.33, 95% CI [0.19, 0.58]). The transport toolkit therefore transfers, the reliability advantage generalizes across independent disorder realizations rather than being tied to one lattice, and the remaining barrier is inference cost rather than coordinate quality.

1. Introduction

A rare event is one that direct simulation almost never observes: the nucleation of a new phase, a protein crossing between conformational basins, or a detector registering a particle that has penetrated many mean free paths of shielding. In each case the probability of the event is itself the quantity of scientific interest, and it is exactly that quantity which unbiased dynamics represents worst. If an event has equilibrium or first-passage probability $p \ll 1$, the relative error of a direct Bernoulli estimator formed from N effectively independent samples follows the elementary result [6, 7]

$$\epsilon_{\text{rel}} = \sqrt{\frac{1-p}{Np}} \simeq (Np)^{-1/2}, \quad p \ll 1, \quad (1)$$

so the sample count required for fixed relative precision grows as $1/p$ and each additional decade of rarity costs an order of magnitude in computation. The failure is therefore structural rather than incidental, and every established remedy responds to it in the same way: not with more samples, but with better-placed ones. This elementary scaling is the origin of a broad family of rare-event techniques, including importance sampling, splitting, transition-path sampling, Forward Flux Sampling (FFS), Adaptive Multilevel Splitting (AMS), and Weighted Ensemble (WE) methods [6, 7, 14, 17, 19, 21, 23]. What each of them requires in practice is a rule for deciding which samples are the better-placed ones.

Two mature traditions answer that question using closely related mathematical objects. In transition-path theory, the committor $h(s)$ is the probability that a trajectory initiated at configuration s reaches a target set before a competing boundary, and it provides an optimal reaction coordinate for that event [24–26]. In classical particle transport, an adjoint

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solution assigns an importance to a phase-space point according to its expected contribution to a detector response; this importance can be used to bias sources, alter transition probabilities, and construct weight windows [8–12]. The formal connection to the Doob h -transform and conditioned Markov processes is well known [27–29]. The contribution of the present work is therefore not to claim a new identity, but to transfer the associated *transport variance-reduction toolkit* to lattice spin dynamics and test, under controlled validation, which parts remain effective.

The transport perspective is attractive because it separates the ideal importance function from the population-control mechanism used when that importance is only approximate. Exact adjoint biasing can in principle produce zero-variance estimators, while weight windows use splitting and roulette to keep statistical weight near a prescribed range. This distinction underlies CADIS and related hybrid deterministic–Monte Carlo variance-reduction methods [11–13]. In the present setting, the same logic motivates using a committor-like quantity to organize a weighted population while retaining the original spin dynamics.

Rare-event population methods have a parallel history outside transport. WE was introduced for biomolecular transition sampling by Huber and Kim, shown to be statistically exact for broad classes of dynamics and binning procedures, and made practical for complex molecular systems by steady-state formulations and mature software [14–16, 18]; FFS and AMS address related first-passage problems through interfaces or adaptive levels [19–21], while transition-path sampling works with reactive-path ensembles and can avoid prescribing a one-dimensional coordinate at all [22, 23]. Machine learning meanwhile offers several routes to the high-dimensional objects these methods need: flow-based and autoregressive samplers for equilibrium distributions [31–34], neural approximations of committor and Feynman–Kac observables from short trajectories [37, 40], learned bias potentials and importance functions [43, 44], and neural Doob dynamics for large deviations [45]. Together these motivate the central practical question of this paper: when the rare event occurs in a high-dimensional spin configuration space, how much is gained by replacing a hand-designed coordinate with a learned, committor-inspired one – measured against a strong physical baseline and with inference cost charged to the result?

The study is therefore organized as a validation ladder rather than a single benchmark. A solvable one-dimensional birth–death chain first verifies the zero-variance limit of exact importance; exact enumeration of a 4×4 Ising lattice then verifies unbiased WE resampling; a larger $L = 20$ Ising calculation tests whether WE genuinely extends a rare magnetization tail at matched spin-update cost; and finally the Edwards–Anderson (EA) spin glass provides a frustrated landscape in which magnetization is a poor coordinate and the low-energy target is a more demanding test for learned configuration-space importance [1–4]. The hierarchy is designed to separate three claims: the exact mathematical importance limit, unbiased weighted population control, and the ability of a learned coordinate to improve access to a genuinely rare target. Throughout, *reliability* (whether a method reaches the event at all, reproducibly) is distinguished from *efficiency* (variance reduction per unit of actual computational cost); the distinction is essential for neural coordinates, because inference overhead is not represented by a spin-update count.

The main findings are positive but bounded. WE is unbiased and extends the accessible Ising tail beyond the direct-sampling floor at matched update cost. In the EA model, a self-consistency-trained neural coordinate significantly improves deep-event discovery over the physical energy coordinate, an advantage that is reproduced across independent disorder realizations rather than tied to one lattice. That improvement is a reliability result: once inference is included in the cost, the same coordinate does not improve the figure of merit. Together, these results identify a transferable theoretical core, a validated route for using learned importance information inside a weight-conserving rare-event sampler, and the specific barrier that a practical implementation must still clear.

2. Theory and Methods

2.1. Adjoint importance, committors, and zero-variance sampling

Let $K(s, s')$ be the transition kernel of a Markov chain and let B denote a target set. For a first-passage event, the forward committor is the probability of reaching B before the competing set A [24, 25]:

$$h(s) = \Pr_s(\tau_B < \tau_A). \quad (2)$$

For a discrete Markov process, this committor satisfies the backward (harmonic) equation in the interior of the state space, a standard result of transition-path theory for Markov jump processes [24–26]:

$$h(s) = \sum_{s'} K(s, s') h(s'), \quad s \notin A \cup B, \quad (3)$$

with boundary values $h = 0$ on A and $h = 1$ on B . Conditioning the dynamics with a positive harmonic function leads to the Doob h -transform, introduced in classical potential theory [27] and used in this form throughout the large-deviation and conditioned-process literature [28, 29]:

$$K_h(s, s') = K(s, s') \frac{h(s')}{h(s)}, \quad (4)$$

where defined. This is the Markov-chain counterpart of adjoint-based importance biasing in transport. With the exact h , successful paths acquire identical total statistical weight, producing the familiar zero-variance ideal.

The relationship between the two importance functions should be stated precisely, since it is weaker than an identity. The committor of Eq. (2) is a probability: the chance of reaching B before A from s , bounded in $[0, 1]$ and defined only once a competing boundary is specified. The transport adjoint is an expected contribution to a specified detector response, carries the units of that response, and is unbounded in general. What the two share is structure rather than definition: both are solutions of a backward equation generated by the same transition operator, both express the value of a state with respect to a terminal objective, and both therefore play the same role when used to tilt or to bin. They coincide as functions in the particular case of an absorbing target with unit detector cross-section and no multiplication, which is the case constructed here; they are not universally the same object, and this paper does not claim they are. What transfers is the backward-conditioning structure and the machinery built on it, not an equivalence of the two quantities in general.

This identity motivates two distinct uses of approximate importance. The first changes transition probabilities directly, which can be fragile when h is inaccurate. The second retains the original dynamics but uses population splitting and merging to control statistical weight. The second strategy is adopted here because weight conservation preserves unbiasedness even when the binning coordinate is imperfect.

2.2. Zero-variance path weighting and the transport correspondence

The zero-variance statement can be seen directly at the path level. Consider a trajectory $\omega = (s_0, s_1, \dots, s_\tau)$ generated under the Doob-transformed kernel until it reaches either A or B . Substituting the Doob transform of Eq. (4) into the path likelihood ratio gives the telescoping identity below, derived here for the discrete first-passage setting from Eqs. (3)–(4); it is the discrete-state counterpart of the classical zero-variance principle of transport importance sampling [8, 9, 12] and of the conditioned-process construction of Doob [27] and Chetrite and Touchette [29]:

$$\prod_{t=0}^{\tau-1} \frac{K(s_t, s_{t+1})}{K_h(s_t, s_{t+1})} = \prod_{t=0}^{\tau-1} \frac{h(s_t)}{h(s_{t+1})} = \frac{h(s_0)}{h(s_\tau)}. \quad (5)$$

On a successful path ending in B , $h(s_\tau) = 1$, so every successful history contributes exactly $h(s_0)$.

The resulting estimator should be stated explicitly, since the zero-variance property depends on how it is constructed. Fix an initial state s_0 with $h(s_0) > 0$ and let the target quantity be the first-passage probability $\pi = h(s_0) = \Pr_{s_0}(\tau_B < \tau_A)$. Generate M independent trajectories under K_h and assign each the likelihood-ratio weight of Eq. (5), giving

$$\hat{\pi} = \frac{1}{M} \sum_{m=1}^M \left[\prod_{t=0}^{\tau_m-1} \frac{K(s_t^{(m)}, s_{t+1}^{(m)})}{K_h(s_t^{(m)}, s_{t+1}^{(m)})} \right] \mathbf{1}_B(s_{\tau_m}^{(m)}). \quad (6)$$

Two properties of K_h make this exact. States with $h = 0$ receive zero transition probability, so the transformed chain never enters A and every trajectory terminates in B ; and on each such trajectory the weight telescopes to $h(s_0)$ by Eq. (5). Every term of the sum therefore equals $h(s_0)$, so $\hat{\pi} = \pi$ identically and $\text{Var}(\hat{\pi}) = 0$ for any $M \geq 1$. The statement is specific to a first-passage probability from a fixed initial state, requires $h(s_0) > 0$, and requires the exact h ; it degrades continuously, and in general non-uniformly, when h is replaced by an approximation. This is the property verified numerically in §3.1.

For an approximate importance function $\tilde{h}$, the exact telescoping property is no longer available. The present implementation therefore separates importance representation from dynamical biasing: the physical Metropolis kernel is retained, while the learned or physical progress coordinate is used only to organize population splitting and merging. This design allows approximate importance information to guide computational effort while weight conservation preserves the target measure.

Table 1 summarizes the operational correspondence used throughout the paper. The mapping is not intended to imply that all transport constructs carry over unchanged. Source biasing, for example, has no literal analogue when the equilibrium spin distribution is fixed and the goal is to sample a stationary tail. The useful transfer is instead structural: an adjoint-like scalar identifies configurations that matter to the response, and a weight-management mechanism concentrates trajectory population in those configurations.

Radiation-transport concept	Markov/spin counterpart	Role in this study
Adjoint importance	Committer / event-conditioned value function	Defines ideal progress toward a rare target
CADIS importance bias	Doob h -transform / tilted dynamics	Verified in the solvable zero-variance demonstration
Weight windows	Weighted splitting and merging	Implemented through WE resampling with conserved statistical weight
Detector response	Rare-event indicator or threshold probability	Magnetization-tail or low-energy probability

Table 1: Transport concepts and their rare-event spin-system counterparts used in this work. The final column states the role tested here.

2.3. Spin models and rare-event targets

For the nearest-neighbor ferromagnetic Ising model, the Hamiltonian is that introduced by Ising [1] and solved exactly in two dimensions by Onsager [2],

$$H(\mathbf{s}) = -J \sum_{\langle ij \rangle} s_i s_j, \quad s_i \in \{-1, +1\}, \quad (7)$$

with periodic boundaries and $J = 1$. The principal rare observable is the absolute magnetization $|m|$. For the spin glass, the corresponding random-bond Hamiltonian is that of Edwards and Anderson [3], in the $\pm J$ form reviewed by Binder and Young [4],

$$H(\mathbf{s}) = - \sum_{\langle ij \rangle} J_{ij} s_i s_j, \quad J_{ij} \in \{-1, +1\}, \quad (8)$$

with one fixed disorder realization per comparison. Quenched disorder and frustration make the low-energy states disordered, so $m \approx 0$ irrespective of energy and magnetization is not an adequate progress coordinate; low energy is used as the rare-event target instead. All reported production comparisons use single-spin Metropolis dynamics [5] at $T = 2.6$ unless stated otherwise.

2.4. Weighted Ensemble estimator and resampling

WE propagates a weighted population under the original, unbiased Metropolis dynamics for a resampling interval τ . After propagation, walkers are partitioned into bins defined by a coordinate $q(s)$. If the total statistical weight in bin b is W_b , resampling redistributes that same W_b among the target number of walkers in the bin. Splitting therefore creates additional trajectories with reduced individual weights, while merging removes redundant trajectories without changing total weight. Because the dynamics themselves are not altered and the resampling operation conserves weight, the coordinate controls efficiency rather than correctness.

For a target set B , WE estimates the target probability from the total statistical weight carried by walkers in B . The estimator takes the weight-conserving form established for WE by Huber and Kim [14], proved statistically exact for broad classes of dynamics and binning procedures by Zhang et al. [15], and used in the steady-state and review formulations of Bhatt et al. [16] and Zuckerman and Chong [17]:

$$\hat{\pi}_t = \frac{\sum_i w_i^{(t)} \mathbf{1}_B(s_i^{(t)})}{\sum_i w_i^{(t)}}, \quad (9)$$

and the reported estimate is averaged over post-burn-in resampling cycles. Three coordinates are compared: absolute magnetization $|m|$, energy per spin E/N , and a learned scalar coordinate $I_\theta(s)$. Magnetization is deliberately retained as a mismatched baseline for the EA spin glass, whereas energy is a natural physical baseline for the low-energy target.

2.5. Rarity calibration and experimental protocol

A fixed rare-event threshold can be either trivial or physically unreachable when lattice size, temperature, or disorder changes. The computational protocol therefore uses a short pilot simulation to locate the most extreme value reached under ordinary dynamics and then places the production threshold beyond that pilot extreme by a small number of discrete lattice steps. For the Ising magnetization target the lattice spacing is $2/L^2$; for the EA energy-per-spin target it is $4/L^2$. The integer “beyond” used below records how many such steps the target is displaced past the pilot extreme. This convention provides an operational depth scale without claiming that equal values of “beyond” correspond to identical probabilities at different system sizes.

The resampling interval is also treated as a physical parameter rather than a bookkeeping choice. Independent diagnostics in the EA model show rapid relaxation of deep configurations: after one Metropolis sweep, the mean energy of a set of deep configurations moves substantially back toward the bulk. Consistent with this, shorter resampling intervals penetrate the deep ladder more effectively than long intervals at comparable cost. The production deep-rare calculation therefore uses $\tau = 2$, while the exact and shallow validation calculations retain their own explicitly stated settings.

2.6. Neural importance coordinate

The learned coordinate is a compact periodic convolutional network acting on the full spin configuration. The implementation uses two circularly padded 3×3 convolutional layers with nonlinear activations, followed by global mean and maximum pooling and a scalar output. This architecture is intentionally small: the research question is whether a configuration-space coordinate can discover useful progress information that is absent from a simple scalar observable, not whether a large neural model can fit the training set.

During development, several training targets were tested. Rollout-based dense surrogates were useful for avoiding the all-zero labels produced by a direct binary committor target in very rare regimes, but such a label is by construction energy-adjacent, since it records the best energy reached in a short walk. Later experiments therefore used objectives shaped by the harmonic self-consistency condition of Eq. (3) itself. This direction sits within a broader neural-committor literature, including direct variational approximation of high-dimensional committors, semigroup formulations that avoid explicit second derivatives, active importance sampling for rare-event-dominated objectives, short-trajectory Feynman–Kac training, and self-consistent committor-enhanced sampling [37–42]. The self-consistency objective used for the discrete MCMC comparison is most directly connected to the recent neural importance-sampling formulation of Kim and Cai [44], in which a learned importance function is coupled to biased transitions and branching. Two practical points follow. First, the self-consistency condition together with the target boundary alone is under-determined, since the constant solution satisfies it everywhere; a second boundary condition anchoring bulk-equilibrium states to a distinct reference value is necessary rather than optional. Second, the present study uses the learned function more conservatively than Kim and Cai, as a WE binning coordinate, so that coordinate quality can be evaluated without introducing likelihood-ratio bias from an inaccurate directional tilt.

A terminological point follows from this design and is maintained throughout. The trained network I_θ is a *learned importance coordinate* whose objective is shaped by the harmonic condition of Eq. (3); it is not claimed to be a quantitatively accurate estimate of the committor h , and no such accuracy is measured here. Establishing that would require a reference committor on a system where one can be computed, which is a separate exercise from the sampling question addressed in this paper. Because I_θ enters only through the binning coordinate, and because weight-conserving resampling is unbiased for any binning, the results below do not depend on how close I_θ is to h : an inaccurate coordinate costs efficiency, never correctness. The word “committor” is therefore reserved for the exact object of §2.1, and I_θ is described as committor-inspired rather than as a committor.

2.7. Computational cost and reporting

Two cost measures are used, and the distinction between them matters for the interpretation of the results. Matched-cost comparisons of sampling reach use the number of attempted spin updates as the common Monte Carlo work unit; this measure is natural for the Ising-tail comparison because both direct sampling and WE use the same Metropolis propagation kernel. It is, however, blind to neural inference, which is performed repeatedly during WE binning and does not appear in a spin-update count. Wall-clock time is therefore also recorded for the learned-coordinate runs and is the measure used whenever a figure of merit is quoted for a neural coordinate. The figure of merit is the cost-normalized

inverse variance standard in transport variance-reduction studies [9, 13],

$$\text{FOM} = \frac{1}{\sigma_{\text{rel}}^2 T}, \quad (10)$$

in which T is the computational cost charged to the calculation; the two cost measures above give the two values of T reported in this work.

Independent replicas are used to quantify sampling variability, and zero-hit replicas are retained explicitly because non-observation is itself meaningful in the deep rare-event regime. The primary learned-coordinate result is reported through event-discovery frequency across independent runs, complemented by Fisher's exact test; cost-normalized efficiency is reported separately in §6.1.

2.8. Statistical analysis and uncertainty quantification

Suppose R independent replicas produce estimates $\hat{\pi}_r$. Following the independent-replica uncertainty reporting recommended for WE studies [17] and implemented in WESTPA [18], the across-replica relative standard deviation is summarized as

$$\sigma_{\text{rel}} = \frac{\text{sd}(\hat{\pi}_1, \dots, \hat{\pi}_R)}{\bar{\pi}}, \quad \bar{\pi} = \frac{1}{R} \sum_{r=1}^R \hat{\pi}_r. \quad (11)$$

For deep targets, however, the most informative quantity can be the fraction of independent runs that observe the event at all. Zero-hit counts are therefore reported explicitly rather than being hidden inside a variance summary.

For the deep-rare neural comparison, event-positive and zero-hit replicas are compared using the exact conditional test for a 2×2 contingency table introduced by Fisher [46], which is appropriate here because the expected counts are small. Where confidence intervals on ratios of figures of merit are quoted, they are obtained by the nonparametric bootstrap [47] in a two-level form that resamples the trained-network axis in the outer loop and the replica axis of each network in the inner loop.

3. Validation

3.1. Exact-importance limit

A biased one-dimensional birth–death chain provides a solvable test. Starting at $s_0 = 1$ with absorbing boundaries at 0 and $N = 15$ and step probability $p = 0.4$ toward the rare boundary, the exact committor is $h(s_0) = 1.144443 \times 10^{-3}$. Direct Monte Carlo gives a relative variance of 2.17×10^{-3} for 4×10^5 histories, in agreement with the Bernoulli prediction $1/(Np)$. Using the exact importance transform reduces the measured relative variance to approximately 10^{-30} , numerically reproducing the zero-variance prediction of Eq. (5). This exact result provides the reference point for the weight-conserving population strategy used in the spin-system calculations.

3.2. Exact-enumeration gate for WE

For the 4×4 Ising model, all 2^{16} configurations can be enumerated. Across the three validation regimes used in the project—above T_c , below T_c with the double-well structure, and the folded $|m|$ distribution—the mean $|\Delta \ln P|$ between WE and exact enumeration ranges from approximately 0.006 to 0.024. The $T = 2.6$ case gives the best of these values, 0.006, and is retained as the representative detailed gate. A separate learned-importance gate gives target-probability ratios of 1.000 and 1.001 relative to exact enumeration when WE is binned by magnetization and energy, respectively. These tests establish the implementation property needed below: resampling is unbiased even when the coordinate is not optimal.

The purpose of this ladder, summarized in Table 2, is methodological. The first test verifies the ideal importance limit, the next two verify unbiased population control against exact answers, and only then is the method used where an exact distribution is unavailable. Consequently, later differences between coordinates are interpreted as efficiency or reliability differences rather than as evidence that one coordinate changes the target measure.

Test	Reference	Outcome
Birth–death importance transform	analytic committor	exact-importance relative variance $\sim 10^{-30}$
4×4 Ising WE distribution	exact enumeration	mean $ \Delta \ln P = 0.006\text{--}0.024$ across three regimes
4×4 learned-coordinate gate	exact target probability	WE ratios 1.000 and 1.001
$L = 20$ Ising tail	matched-update naive MC	WE resolves one additional rare-tail bin to 2.4×10^{-7}

Table 2: Validation ladder used before interpreting the larger rare-event calculations. Each row validates a different component of the computational framework.

4. Rare-Event Reach in the Ising Ferromagnet

4.1. WE extends the Ising tail

At $L = 20$ and matched update cost ($\sim 3.4 \times 10^8$ spin updates), naive sampling and WE agree throughout the well-sampled region, and their behavior separates only in the far tail. Naive sampling reaches $|m| = 0.886$ at $P \approx 5.9 \times 10^{-6}$ but records no samples in the next tail bin, whereas WE estimates $P \approx 2.4 \times 10^{-7}$ at $|m| = 0.932$. The claim supported by this comparison is that WE reaches a tail bin approximately 24 times below the direct-sampling floor under the matched update budget, although the deepest estimate is itself statistically noisy: it carries an across-replica relative standard deviation of approximately 2.0 over four replicas. The robust content is therefore that WE returns an estimate where direct sampling returns none at all, rather than that the value 2.4×10^{-7} is precise. Expressed as sampling reach, the lowest probability resolved under the common budget falls from 5.9×10^{-6} to 2.4×10^{-7} , an extension of approximately 1.4 orders of magnitude, and this is the quantity the rest of the paper depends on.

4.2. Calibration of the deep-rare spin-glass regime

The neural-coordinate experiment is placed deliberately beyond the range where direct sampling is consistently informative. The calibrated EA thresholds at $L = 16$ become progressively less accessible under the same direct-sampling budget. Table 3 summarizes this depth calibration and identifies the access-dominated regime used for the learned-coordinate test.

Beyond	$\hat{\pi}$	relative s.d.	zero-hit replicas
1	5.29×10^{-5}	0.23	0/10
2	5.48×10^{-6}	0.46	0/10
3	7.14×10^{-7}	1.61	7/10
4	0	—	10/10

Table 3: Depth calibration for the EA model at $L = 16$ using 10 independent direct-sampling replicas. The integer “beyond” indexes successively deeper discrete energy thresholds.

The beyond= 4 threshold is therefore used as a stress test of rare-event access. This choice makes the learned-coordinate comparison a direct test of whether a configuration-space importance map can organize the weighted population so that trajectories continue to enter a region that is essentially absent from the corresponding direct Monte Carlo sample.

5. Learned Importance in the Deep-Rare Spin Glass

At beyond= 4 ($E/N \leq -1.119$), direct Monte Carlo is effectively blind under the calibrated budget. The WE budget was increased to 80 bins, 40 walkers per occupied bin, 3000 iterations, and $\tau = 2$. Three independently trained self-consistency networks were then compared with the same physical-coordinate baselines. Using independently trained networks is important because the objective is to establish a reproducible coordinate-level effect rather than a favorable outcome from one particular initialization.

Method	event-positive replicas	FOM _{MC}	ratio vs. WE[E]
naive	1/30	2.48×10^{-10a}	—
WE[m]	5/30	1.04×10^{-10}	0.62
WE[E]	12/30	1.69×10^{-10}	1.00
WE[I_θ], seed 0	19/30	2.18×10^{-10}	1.29
WE[I_θ], seed 1	20/30	1.81×10^{-10}	1.07
WE[I_θ], seed 2	21/30	1.77×10^{-10}	1.04

^aNot a meaningful figure of merit: driven by a single event-positive replica out of 30 (relative s.d. = 5.5); reported for completeness only. The underlying $\hat{\pi}$ estimates for the weight-window methods remain individually noisy in this regime, with relative s.d. between 1.2 and 1.9 at 30 replicas.

Table 4: Deep-rare EA comparison at $L = 16$, beyond= 4. Each learned-coordinate row is a separately trained network evaluated over 30 replicas. Figures of merit are quoted on the spin-update cost measure of §2.7; the wall-clock correction is applied in §6.1.

The primary evidence is the network-level replication. Each of the three independently trained networks exceeds the energy coordinate on its own: discovery rates of 63%, 67%, and 70% (19/30, 20/30, 21/30) against 40% (12/30) for WE[E] (Figure 1). The effect is therefore reproduced three times from independent initializations, with no overlap in direction, which is the property that matters for a coordinate-level claim.

Pooling those runs gives 60 event-positive replicas out of 90 (66.7%) against a naive reading of 12/30 (40.0%), and Fisher’s exact test on the resulting table gives $p = 0.017$. That WE[E] figure is a single fixed 30-replica draw, however, reused as the comparison point for all three networks; a further 60 independent replicas (disjoint random seeds, same configuration) give a better-calibrated estimate of its true discovery rate, 44/90 (48.9%) pooled, close to but above the original draw, which happened to land on the low side by ordinary sampling variation. Repeating the test against this balanced baseline gives 66.7% versus 48.9%, $p = 0.023$ – a smaller margin than the naive comparison but still significant, and the value that should be quoted going forward. This pooled test is offered as supporting evidence rather than as the sole statistical basis, because the 90 learned-coordinate replicas are grouped within three trained networks and are not fully exchangeable. The grouping appears benign here, since the three per-network rates lie within 7 percentage points of one another and a network-level effect large enough to invalidate pooling would be expected to produce visible heterogeneity; but the replication across networks, not the pooled p -value, is what the conclusion rests on. This is the central within-disorder learned-coordinate result: the learned importance map increases the probability that a finite-budget weighted-ensemble run reaches the deep target. §5.1 extends this replication axis from network initialization to independent disorder realizations.

The comparison is especially informative because energy is a strong physical baseline for this task. The target itself is defined by low energy, so an improvement over WE[E] cannot be attributed to comparing against an irrelevant coordinate. The network instead receives the full spin configuration and can distinguish configurations with similar instantaneous energy but different local structure and different dynamical prospects of progressing toward the target. This is also the mechanistic reason to prefer a self-consistency objective over the rollout surrogate of §2.6: two configurations at identical energy can have very different committor values on a frustrated landscape, and an energy-adjacent label cannot represent that difference. The present experiment does not attempt to assign a unique microscopic interpretation to the learned representation; it establishes that the representation contains useful progress information beyond the scalar energy coordinate.

The role of the learned function is deliberately conservative. It is used to define the WE partition rather than to replace the physical transition kernel. Because the split/merge operation conserves total statistical weight, the estimator remains unbiased for any binning coordinate. The learned model therefore influences how computational effort is distributed across configuration space without changing the equilibrium target measure. This separation is attractive for extending neural rare-event methods to more complex spin systems, where direct use of an approximate committor inside a likelihood-ratio tilt would require stronger control of approximation error.

5.1. Generalization across disorder realizations

The result above establishes reproducibility across network initializations on one J_{ij} sample; it does not by itself establish that the effect is a property of the method rather than of that particular disorder realization. Two further independent EA disorder realizations were therefore tested at the same $L = 16$, $T = 2.6$, beyond= 4 configuration,

one trained network each – the network-initialization axis having already been shown not to be the dominant source of variance, a single network per realization is the appropriate allocation of computational effort here rather than a further three-network ensemble per realization.

Table 5 reports each realization separately, since a pooled number alone cannot show whether one realization drives the effect. No single realization reaches significance on its own at $n = 30$: the original realization, taken as a single-network comparison rather than the three-network pooled result above, is itself not significant (19/30 versus 12/30, $p = 0.120$), so this is the expected pattern at this replica count rather than a weakening of the result. The third realization has a substantially easier calibrated target for its particular disorder (29/30 and 26/30, both near ceiling), which is why its own test is the least informative of the three – a ceiling effect, not evidence against the finding.

Disorder realization	WE[I_θ]	WE[E]	Fisher p
12345 (original)	19/30 (63.3%)	12/30 (40.0%)	0.120
23456	15/30 (50.0%)	8/30 (26.7%)	0.110
45678	29/30 (96.7%)	26/30 (86.7%)	0.353
Pooled (3 realizations)	63/90 (70.0%)	46/90 (51.1%)	0.014

Table 5: Deep-rare EA comparison repeated across three independent disorder realizations, one trained network each, beyond= 4, 30 replicas per method. Realization 12345 is the single-network reading of the comparison in Table 4 (seed 0), not the three-network pooled result.

Deep-rare EA discovery rate ($L = 16$, beyond = 4): error bars are Wilson 95% intervals; diamonds in (a) are the 3 individual trained networks

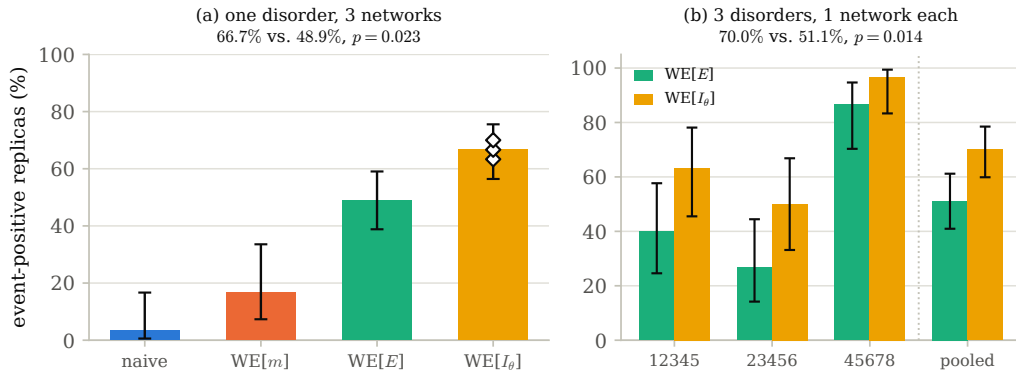


Figure 1: The learned importance coordinate reaches the deep target more reliably than the physical energy coordinate, in the regime where direct sampling cannot compete. Error bars are Wilson 95% intervals. (a) Discovery rate at beyond= 4 (disorder realization 12345, $L = 16$, $T = 2.6$): open diamonds are the three independently trained networks (Table 4); WE[E] uses the balanced 90-replica baseline (48.9%; the original unbalanced 30-replica draw gave 40.0%). (b) The same comparison repeated across three independent disorder realizations, one trained network each (Table 5), with the pooled result at right.

Pooled across all three independent realizations, one network each (90 replicas per method):

$$\text{WE}[I_\theta] : 63/90 (70.0\%) \quad \text{versus} \quad \text{WE}[E] : 46/90 (51.1\%), \quad p = 0.014. \quad (12)$$

This is a different and stronger pooling axis than the network-seed result above: rather than asking whether the effect is robust to network initialization on one lattice, it asks whether the effect is a property of the method across the disorder ensemble. Wall-clock cost ratios for the two new realizations (4.45 and 4.97 relative to WE[E]) are consistent with every other measurement in this study (§6.1), so the efficiency conclusion generalizes together with the reliability one.

A methodological note is worth recording. A third candidate disorder realization was attempted and abandoned before completion: its real-depth training-data collection pool returned no states in the bulk-equilibrium reference region at all, so the self-consistency objective's second boundary condition (§2.6) could never fire and training collapsed toward the trivial constant solution, a failure mode the training code detects and flags explicitly. This was verified to be a property of that particular J_{ij} sample's dynamics under the collection protocol, not a general problem: two replacement realizations were each checked with a short collection-only diagnostic before committing to the full run, and the one reported here was healthy by that check. Screening a new disorder realization this way before full training is accordingly recommended practice for any extension of this result.

6. Discussion

The results support a direct progression from transport theory to rare-event sampling in spin systems. The first level is mathematical: for first-passage problems, the committor is the harmonic importance function and the Doob transform supplies the zero-variance ideal. The birth–death demonstration verifies this limit numerically and provides a transparent reference against which approximate importance strategies can be interpreted.

The second level is algorithmic. Weight-conserving WE resampling transfers the population-control logic of transport weight windows without requiring the importance coordinate to be exact. Exact enumeration on the 4×4 Ising model confirms that the target measure is recovered across the tested regimes, and the $L = 20$ tail calculation shows why population control becomes valuable in practice: once the direct sample reaches its floor, the weighted population continues to populate states with probabilities more than an order of magnitude smaller.

The third level is representational. In the EA spin glass, magnetization is not a useful descriptor of a low-energy rare event, while energy is informative but compresses the configuration to one scalar. A learned committor-like coordinate can use the full lattice pattern to rank configurations by their prospects of progressing toward the target. The significant increase from a balanced 48.9% to 66.7% event-positive replicas demonstrates that this additional information can improve deep-event access. Because the effect is reproduced by three independently trained networks on one realization, and again by pooling across three independent disorder realizations (51.1% to 70.0%, $p = 0.014$, §5.1), the result is not tied to one isolated initialization or to one quenched-disorder sample.

6.1. Reliability is improved; cost-normalized efficiency is not

The improvement reported in §5 is a reliability result, and it should not be read as a claim of computational speedup. On the spin-update cost measure, the three trained networks give figure-of-merit ratios of 1.29, 1.07, and 1.04 against $\text{WE}[E]$ (Table 4); a two-level bootstrap over networks and replicas gives a geometric-mean ratio of 1.16 with 95% interval [0.65, 2.12], which does not exclude parity. That measure is in any case incomplete, because it does not charge for the network evaluations performed at every resampling step.

Direct measurement removes the need to extrapolate that omission from a proxy. All three deep-rare networks were rerun end-to-end with wall-clock time recorded as a first-class output for every method (not reconstructed from log timestamps), reusing the deterministic trained weights so the underlying $\hat{\pi}$ estimates are unchanged. Measured wall-clock ratios of $\text{WE}[I_\theta]$ to $\text{WE}[E]$ were 4.65, 4.01, and 4.54 for the three networks – consistent in direction with, though somewhat larger than, an earlier profiling-based estimate of 2.9–3.4 for a single network obtained at a shallower configuration. The $\text{WE}[E]$ baseline itself was also expanded from the original fixed 30 replicas to a balanced 90 (§5); the wall-clock-fair figure of merit uses this larger pool only to obtain a better-calibrated estimate of $\text{WE}[E]$'s native-replica-count precision (bootstrap-resampling 30-replica blocks from the pool of 90, at $\text{WE}[E]$'s own measured wall-clock cost for 30 replicas), rather than charging its full expanded wall-clock time against $\text{WE}[I_\theta]$'s native budget, which would inflate $\text{WE}[E]$'s effective cost and bias the ratio. The resulting two-level-bootstrapped, wall-clock-fair figure-of-merit ratio is

$$\text{FOM}_{\text{wall}}(\text{WE}[I_\theta]) / \text{FOM}_{\text{wall}}(\text{WE}[E]) = 0.33, \quad 95\% \text{ CI} = [0.19, 0.58], \quad (13)$$

below parity and excluding 1. This directly measured value agrees in direction and order of magnitude with an earlier estimate obtained by rescaling the spin-update-cost ratio with a profiling-based timing factor from a shallower configuration (0.39, [0.22, 0.72]), which is a useful cross-check of that method, but Eq. (13) is the value that should be quoted going forward. The corresponding value for the five-network ensemble, approximately 0.087, is obtained at the shallower configuration where the ensemble was characterized and is reported as an indication of how the penalty scales with network count, not as a second deep-rare measurement.

Table 6 consolidates the reliability and cost results so that the trade-off can be read in one place.

Method	networks	rel. wall time	event-positive	FOM _{MC} ratio	FOM _{wall} ratio
WE[E] ^a	0	1.00	44/90 ^c	1.00	1.00
WE[I_θ] ^a	1	4.0–4.7	60/90	1.16 [0.65, 2.12]	0.33 [0.19, 0.58]
WE[I_θ] ^b	5	10.7–10.9	—	0.74 [0.43, 1.32]	0.087 [0.051, 0.155]

^aDeep-rare configuration of §5 (beyond= 4, 30 replicas per network), directly measured (§6.1). ^bEnsemble characterization at the shallower configuration (beyond= 1, $\tau = 2$, 10 replicas per run, five independent base seeds), where every method reaches the target in every replica so that event-positive counts are uninformative; this row is still extrapolated from that shallower configuration, not directly measured at beyond= 4.

^cBalanced 90-replica baseline (§5); the naive fixed-30 draw gave 12/30.

Table 6: Reliability and cost of the learned coordinate relative to the energy coordinate. Wall time is relative to WE[E] under the same configuration. FOM_{MC} ratios are geometric means with two-level bootstrap 95% intervals on the spin-update cost measure; FOM_{wall} ratios are directly measured (single network) or extrapolated (ensemble), see notes. Note that the two learned-coordinate rows come from different configurations and are not a like-for-like comparison with each other.

Two properties of these values should be stated plainly. The single-network deep-rare figure is now a direct measurement rather than an extrapolation, and its interval propagates sampling uncertainty in the $\hat{\pi}$ estimates on both sides of the ratio; the five-network ensemble figure is still obtained by extrapolating a timing factor measured at a shallower configuration, and its interval does not include uncertainty in that factor itself. The conclusion that survives these caveats is qualitative but robust to the details: a learned coordinate can be dynamically better and simultaneously worse per unit of computer time, and increasing neural capacity moves the second property in the wrong direction faster than it improves the first. A dedicated efficiency study with timing recorded as a first-class output throughout, and with inference cost reduced by caching or batching, is the natural sequel to the present work; the single-network figure now meets that bar, the ensemble figure does not yet.

6.2. Relation to neighboring rare-event methods

The present implementation is one member of a broader splitting and path-sampling family, and the relevant contrast is mechanistic. FFS and AMS advance trajectories through fixed interfaces or adaptively chosen levels [19–21], and transition-path sampling works directly with reactive-path ensembles, which frees it from a fixed scalar coordinate but targets different estimators than the stationary tail probabilities studied here [22, 23, 30]. WE differs in maintaining a continuously weighted population and resampling within bins, so that the bins influence variance but not the target measure [14, 15, 17]. What the transport analogy adds is a principled target for choosing those bins: CADIS couples an importance map to source biasing and weight windows [11–13], and the present implementation adopts the population-control half of that logic without altering the Metropolis kernel.

Against the neural rare-event literature, the distinction is where the learned object is inserted. Flow-based and autoregressive samplers learn global equilibrium structure [31–34]; neural committor and Feynman–Kac methods target dynamical progress toward a specific event [37, 38, 40]; and active or self-consistent committor schemes concentrate training and sampling effort in transition-relevant regions [39, 41, 42]. The framework of Kim and Cai is closest in spirit, combining a learned event-specific importance function with biased transitions and branching [44]: in the language of §2.1, their importance function plays the role of h , their modified transitions the role of the tilt, and their branching the role of splitting. All of these place the learned function inside the dynamics. The present approach places it only in the binning, which preserves unbiasedness through weight conservation and makes coordinate quality measurable without confounding it with likelihood-ratio error.

6.3. Scope and reproducibility

The principal learned-coordinate experiment is a proof of concept, and its scope should be read literally. It uses three two-dimensional EA disorder realizations, one temperature ($T = 2.6$), one lattice size ($L = 16$), one threshold depth (beyond= 4), and a compact periodic convolutional network. Replication in this study is across trained networks, disorder realizations, and simulation replicas, not yet across lattice size or temperature. The result therefore establishes that a learned configuration-space coordinate *can* carry progress information beyond energy on a frustrated landscape, and does so reproducibly across network initializations (§5) and across three independent disorder realizations (§5.1);

it does not establish generality across lattice size or temperature, and the three-realization sample is not large enough to characterize how the effect's magnitude varies with disorder – one of the three realizations tested presented a substantially easier calibrated target than the other two. A larger disorder sample, to tighten the pooled confidence interval and better characterize this variation, and extensions in L and T , are the natural next steps, and are more expensive than the disorder-axis test already completed. Exact enumeration, the solvable birth–death reference, fixed-seed reproducibility checks, and explicit reporting of zero-hit replicas provide complementary safeguards for the numerical interpretation. One further caution follows from the project's own experience: because outcomes vary appreciably between training initializations at this scale, a single training run is not evidence about a learned coordinate in either direction, and seed-resolved reporting should be treated as a requirement rather than a refinement. All learned-coordinate results reported here are accordingly stated across independently trained networks. A second, disorder-specific caution also follows from this project's experience: not every disorder realization is trainable under the self-consistency objective's collection protocol, since the second boundary condition can fail to fire for a particular J_{ij} sample's dynamics (§5.1); a brief collection-only diagnostic before committing to full training is accordingly recommended when extending this result to further realizations.

The next natural extension is to test the same learned-coordinate construction across lattice sizes and lower-temperature regimes where the separation between common and rare configurations becomes stronger, and across a larger sample of disorder realizations. Architectures that encode spin symmetries or local frustration explicitly may further improve the quality of the learned importance map. The same framework can also be coupled to adaptive bins or milestone definitions, provided the underlying split/merge step remains weight conserving.

7. Conclusions

Adjoint transport theory provides a useful organizing principle for rare-event sampling in lattice spin systems. The committor supplies the event-specific importance function, the exact Doob transform defines the zero-variance ideal, and weight-conserving population control provides a practical way to exploit approximate importance without changing the target measure. These elements were validated sequentially from a solvable birth–death process to exact enumeration of the Ising model.

At system scale, WE reaches an $L = 20$ Ising magnetization-tail bin approximately 24 times below the direct-sampling floor near 5.9×10^{-6} at matched spin-update cost, returning a (noisy) estimate of 2.4×10^{-7} where direct sampling returns none. In the frustrated EA model, the learned configuration-space importance coordinate produces the main new result: pooled across three independent disorder realizations with one trained network each, it yields 63 event-positive replicas out of 90 against 46/90 for the physical energy coordinate (70.0% versus 51.1%, Fisher exact $p = 0.014$); a complementary pooling of three independently trained networks on a single realization against a matched-size baseline gives 66.7% versus 48.9% ($p = 0.023$). The result shows that a learned progress coordinate can extract dynamical information that is not fully represented by a scalar physical observable, and that this advantage is reproducible both across network initializations and across independent disorder realizations – a property of the method rather than of one particular quenched-disorder sample. A larger disorder sample would further tighten this estimate and is the natural next step.

That advantage is presently one of reliability rather than efficiency. Once repeated network inference is directly measured and charged against the result, the learned coordinate sits below the energy coordinate in wall-clock figure of merit by a factor of approximately three for a single network (0.33, 95% CI [0.19, 0.58], excluding parity) and by an order of magnitude for a five-network ensemble. Reducing that inference cost, rather than improving the coordinate further, is therefore the binding constraint on making learned importance practical at this scale, and is the specific target of the follow-up study.

The resulting framework connects classical transport variance reduction, WE population sampling, and neural importance learning in a single validated workflow. It is directly extensible to larger spin systems and to other rare-event problems in which a physically obvious one-dimensional coordinate is insufficient.

Data and Code Availability

All code, input data, and result files required to reproduce the results reported here are publicly archived [48]. The deposit contains the Weighted Ensemble implementation and production drivers, the training code for both objectives, the quenched disorder realizations used for every Edwards–Anderson result, per-replica $\hat{x}$ arrays for every table and

figure, the exact-enumeration and birth–death validation outputs, the trained network weights for the three within-disorder deep-rare networks and the two additional disorder-realization networks of §5.1, the raw wall-clock logs underlying the cost analysis of §6.1, and a manifest mapping each published number to the file that produced it. An environment specification pins package versions, and all published results are CPU-deterministic under the recorded seeds. Results not reported in the main text, including the seed sweep and ensemble runs of §6.1, are retained in full in the same deposit and described in the Supplementary Material.

Acknowledgements

The author gratefully acknowledges the support of Fahad Bin Sultan University, Tabuk, Saudi Arabia, including access to the computational environment used for the numerical experiments.

CRedit authorship contribution statement

M.Y. Alzaatreh: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing.

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